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PRACTICE Series

NEET UG 2027

Most Important Questions: 120 PYQs with Solutions

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NEET 2027:

120 Most Important Questions From Previous Years

120 real questions from official NEET/AIPMT papers (2015–2026),
hand-picked for repeat value, with answers and explanations

Sourced from official NEET (UG) & AIPMT question papers, 2015–2026

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About the Exam

NEET (UG) — the National Eligibility cum Entrance Test for Undergraduates — is the single national-level entrance examination for admission to MBBS, BDS, AYUSH, BSc Nursing and other undergraduate medical courses in India. It is conducted by the National Testing Agency (NTA) and is the gateway that over two million aspirants attempt every year.

The exam covers three subjects — Physics, Chemistry and Biology (Botany and Zoology) — through 180 compulsory multiple-choice questions carrying a total of 720 marks. The questions are split 45 each across Physics, Chemistry, Botany and Zoology, with +4 for every correct answer, -1 for every incorrect one, and 0 for anything left unattempted. Biology (Botany + Zoology together) accounts for 360 of those marks, making it the single highest-weightage section. The standing duration is 3 hours (180 minutes).

Two pattern updates are worth knowing. First, the optional 'Section B' introduced during the pandemic — where candidates chose 180 questions out of 200 — has been discontinued; every question is now compulsory, so any older practice material built on the 200-question format is out of date. Second, the Education Ministry has confirmed that NEET (UG) will move from the pen-and-paper OMR format to a Computer-Based Test (CBT) from the 2027 cycle onwards. NTA's full information bulletin for 2027 is still awaited, so candidates should confirm final details — exam date, shift timings, centres — on the official NTA NEET portal before relying on them.

Because the syllabus is vast and largely NCERT-based, one of the highest-value habits a serious NEET aspirant can build is solving previous years' official question papers — not only to practice under real exam conditions, but to notice which topics, phrasings and 'traps' NTA keeps returning to year after year.

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About This Ebook

This ebook brings together 120 real questions taken directly from official NEET (UG), AIPMT and Re-NEET question papers spanning 2015 to 2026 — ten questions from each of the last ten NEET/AIPMT years (2015–2024), plus ten each from NEET 2025 and the June 2026 Re-NEET exam, spread across Physics, Chemistry and Biology in roughly the same proportion they carry in the actual exam.

A quick note on 2026: the NEET (UG) 2026 exam originally held on 3 May 2026 was cancelled by the NTA following a paper-leak controversy, and a fresh exam — Re-NEET 2026 — was conducted on 21 June 2026. Its result is the one that stands for admissions, so the '2026' questions in this ebook are drawn from that Re-NEET 2026 paper.

Every question here was asked in an actual NTA-conducted exam. None of the questions in this ebook have been invented — each has been sourced from official previous-year papers and cross-checked against a public NEET/AIPMT archive. The four options presented for each question, the marking of the correct one, and the explanation underneath have all been written independently for this ebook, in plain language, so you understand the 'why' and not just the 'what'.

Use it as a focused, high-density revision sprint — attempt each question yourself first, then check your reasoning against the explanation. Because the same handful of concepts resurface across multiple years (you'll notice a few repeat almost word-for-word), working through all 120 will also train you to spot NTA's favourite patterns.

All the best for NEET (UG) 2027!

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How These 120 Questions Were Chosen

Not every question from twelve NEET/AIPMT papers deserves a spot in a 'most important' list. Each question selected for this ebook satisfies at least one — often more than one — of the following criteria:

Repeated / high-frequency topics — Concepts that NTA has tested in some form across multiple years — e.g. C4 photosynthesis, restriction enzymes, heterosporous pteridophytes, Hardy-Weinberg equilibrium, or ABO blood grouping. If a topic keeps reappearing, it is a safe bet for the next attempt too.

Conceptually tricky / easily confused — Questions that hinge on a subtle distinction most students get backwards on a first read — such as which pair of statements is 'incorrect', or apoenzyme vs prosthetic group, or ureotelic vs uricotelic excretion.

Multi-concept / integrative questions — Questions that force you to connect two chapters at once — for example, linking cell biology (organelle structure) with genetics (DNA packaging), or physiology (ECG waves) with basic physics of electrical signals.

High NCERT-diagram or definition dependence — Straight-from-the-textbook questions (a named scientist, a named experiment, a single defining line) that are trivial once known but costly if skipped during revision.

Numerically or logically layered — Short calculation or reasoning-chain questions (Mendelian ratios, ATP/NADPH accounting, recombination frequency) where a single conceptual slip changes the final answer.

Within each year, the ten questions selected also roughly mirror the real exam's subject-wise weightage, with Biology forming the largest share, followed by Chemistry and Physics.

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NEET 2015 — Previous Year Questions

Q1. [NEET 2015 — Biology — Body Fluids & Circulation]

During the cardiac cycle, which event produces the second heart sound heard through a stethoscope?

(A) Closure of the atrioventricular (tricuspid/bicuspid) valves

(B) Closure of the semilunar valves (aortic and pulmonary)

(C) Opening of the semilunar valves

(D) Contraction of the ventricular muscle

Correct Answer: (B) Closure of the semilunar valves (aortic and pulmonary)

Explanation: The first heart sound (lub) marks the closing of the atrioventricular valves at the start of systole. The second sound (dub) is produced when the semilunar valves snap shut as the ventricles relax, preventing backflow of blood from the aorta and pulmonary artery.

Q2. [NEET 2015 — Biology — Breathing and Exchange of Gases]

Which pulmonary disorder is characterised by a drastic reduction in alveolar surface area, caused by damage to the alveolar walls?

(A) Asthma

(B) Occupational pneumoconiosis

(C) Emphysema

(D) Pleurisy

Correct Answer: (C) Emphysema

Explanation: In emphysema, alveolar walls break down and fuse into larger, fewer air sacs, cutting the total surface area available for O2/CO2 exchange. It is commonly caused by long-term smoking.

Q3. [NEET 2015 — Biology — Structural Organisation in Animals]

What is the main physiological role of chromatophores in animals?

(A) Digestion of food particles

(B) Colour change and camouflage

(C) Production of digestive enzymes

(D) Conduction of nerve impulses

Correct Answer: (B) Colour change and camouflage

Explanation: Chromatophores are pigment-containing cells (found in amphibians, fish, reptiles, cephalopods) that expand or contract to change an animal's skin colour, aiding camouflage, communication, or thermoregulation.

Q4. [NEET 2015 — Biology — Sexual Reproduction in Flowering Plants]

What does the liquid 'coconut water' found inside a tender coconut actually represent, structurally?

(A) Free-nuclear liquid endosperm

(B) Liquid nucellus tissue

(C) Degenerated embryo sac fluid

(D) Cellular endosperm in dissolved form

Correct Answer: (A) Free-nuclear liquid endosperm

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Explanation: In coconut, the endosperm develops in the free-nuclear form first — numerous nuclei float in the liquid without immediate cell wall formation — before later becoming cellular (the white 'meat' of a mature coconut).

Q5. [NEET 2015 — Biology — Morphology of Flowering Plants]

The wheat grain's embryo has one large, shield-shaped cotyledon. What is this structure called?

(A) Coleoptile

(B) Scutellum

(C) Coleorhiza

(D) Plumule

Correct Answer: (B) Scutellum

Explanation: In grasses (family Poaceae) like wheat and maize, the single cotyledon of the monocot embryo is modified into a shield-shaped scutellum, which absorbs nutrients from the endosperm during germination.

Q6. [NEET 2015 — Biology — Diversity in Living World]

The phenomenon of an organism alternating between an asexually reproducing polyp phase and a sexually reproducing medusa phase is called what?

(A) Polymorphism

(B) Metamerism

(C) Metagenesis

(D) Neoteny

Correct Answer: (C) Metagenesis

Explanation: Metagenesis (alternation of generations within the same life cycle, without a change in ploidy) is a defining feature of many Coelenterates/Cnidarians, classically illustrated using Obelia in NCERT.

Q7. [NEET 2015 — Biology — Biotechnology: Principles and Processes]

The ability to cut DNA at specific locations became possible with the discovery of which class of enzymes?

(A) DNA ligases

(B) DNA polymerases

(C) Restriction endonucleases

(D) RNA primases

Correct Answer: (C) Restriction endonucleases

Explanation: Restriction endonucleases recognise specific palindromic sequences and cut both DNA strands at defined points, making them the foundational 'molecular scissors' of recombinant DNA technology.

Q8. [NEET 2015 — Biology — Biotechnology and Its Applications]

Golden Rice is a genetically modified crop in which the inserted gene enables biosynthesis of which compound?

(A) Beta-carotene (Vitamin A precursor)

(B) Lysine (an essential amino acid)

(C) Vitamin C

(D) Omega-3 fatty acid

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Correct Answer: (A) Beta-carotene (Vitamin A precursor)

Explanation: Golden Rice was engineered to accumulate beta-carotene in its edible grain (rice endosperm does not normally make it), to help address Vitamin A deficiency in rice-dependent populations.

Q9. [NEET 2015 — Chemistry — Coordination Compounds]

What is the hybridisation of nickel in the complex ion [Ni(CN)4]2-?

(A) sp3

(B) dsp2

(C) sp3d

(D) d2sp3

Correct Answer: (B) dsp2

Explanation: CN- is a strong-field ligand; it pairs up Ni2+'s 3d electrons, freeing one d-orbital to combine with one s- and two p-orbitals, giving dsp2 hybridisation and a square planar, diamagnetic complex.

Q10. [NEET 2015 — Chemistry — Organic Compounds Containing Nitrogen]

How many structural isomers are possible for the molecular formula C3H9N?

(A) Two

(B) Three

(C) Four

(D) Five

Correct Answer: (C) Four

Explanation: C3H9N corresponds to an amine with 3 carbons: 1-propanamine, 2-propanamine (isopropylamine), N-methylethanamine, and trimethylamine — four structurally distinct isomers in total.

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NEET 2016 — Previous Year Questions

Q11. [NEET 2016 — Biology — Photosynthesis]

The discovery of the Emerson Enhancement Effect and the 'Red Drop' phenomenon helped establish what key idea about photosynthesis?

(A) Chlorophyll a is the only pigment needed

(B) Two distinct photosystems (PS I and PS II) act together

(C) Photosynthesis occurs only at night

(D) CO_2 fixation requires no light at all

Correct Answer: (B) Two distinct photosystems (PS I and PS II) act together

Explanation: Emerson found photosynthetic efficiency dropped sharply beyond 680 nm (red drop) but was restored by combining far-red with shorter-wavelength light (enhancement effect) — evidence that two photosystems cooperate in light reactions.

Q12. [NEET 2016 — Biology — Cell Structure and Function]

Which class of pigments, found dissolved in plant cell vacuoles, is water-soluble?

(A) Chlorophylls

(B) Carotenoids

(C) Anthocyanins

(D) Xanthophylls

Correct Answer: (C) Anthocyanins

Explanation: Unlike chlorophylls and carotenoids (fat-soluble, located in plastids), anthocyanins are water-soluble flavonoid pigments stored in the cell vacuole, responsible for many red, purple and blue plant colours.

Q13. [NEET 2016 — Biology — Sexual Reproduction in Flowering Plants]

What is the term for seed formation in flowering plants that occurs without fertilisation?

(A) Parthenocarpy

(B) Apomixis

(C) Vivipary

(D) Polyembryony

Correct Answer: (B) Apomixis

Explanation: Apomixis produces seeds asexually — often via nucellar embryony — mimicking normal sexual reproduction in outward appearance. It's commercially valuable for fixing hybrid vigour across generations.

Q14. [NEET 2016 — Biology — Biotechnology: Principles and Processes]

The thermostable Taq DNA polymerase, essential for PCR, is obtained from which organism?

(A) Escherichia coli

(B) Thermus aquaticus

(C) Agrobacterium tumefaciens

(D) Bacillus thuringiensis

Correct Answer: (B) Thermus aquaticus

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Explanation: Taq polymerase is isolated from Thermus aquaticus, a bacterium living in hot springs; its heat-stability lets it survive the ~95°C denaturation step repeated in every PCR cycle.

Q15. [NEET 2016 — Biology — Biodiversity and Conservation]

What is considered the single most important cause driving animal and plant species to extinction?

(A) Over-exploitation

(B) Habitat loss and fragmentation

(C) Co-extinction

(D) Alien species invasion

Correct Answer: (B) Habitat loss and fragmentation

Explanation: Among the 'Evil Quartet' of extinction causes, habitat loss and fragmentation is regarded as the most significant — and this exact fact has repeated across multiple NEET years.

Q16. [NEET 2016 — Biology — Anatomy of Flowering Plants]

In which tissue/region of the root are Casparian strips found?

(A) Epidermis

(B) Cortex

(C) Endodermis

(D) Pericycle

Correct Answer: (C) Endodermis

Explanation: Casparian strips are band-like deposits of suberin on the radial and transverse walls of endodermal cells; they force water to pass through the endodermal cell membrane, enabling selective absorption.

Q17. [NEET 2016 — Biology — Ecosystem]

Who coined the term 'ecosystem'?

(A) Eugene Odum

(B) A. G. Tansley

(C) Ernst Haeckel

(D) Charles Elton

Correct Answer: (B) A. G. Tansley

Explanation: The British ecologist Arthur Tansley introduced the term 'ecosystem' in 1935 to describe the interacting system of living organisms and their physical environment.

Q18. [NEET 2016 — Biology — Principles of Inheritance and Variation]

In a cross between a true-breeding tall pea plant and a true-breeding dwarf pea plant, what phenotypic ratio appears in the F2 generation after selfing F1?

(A) 1 : 1

(B) 1 : 2 : 1

(C) 3 : 1

(D) 9 : 3 : 3 : 1

Correct Answer: (C) 3 : 1

Explanation: F1 is all Tt (tall, dominant). Selfing F1 (Tt x Tt) gives the familiar 3 tall : 1 dwarf phenotypic ratio in F2 — one of the most repeated 'safe mark' calculations in NEET Genetics.

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Q19. [NEET 2016 — Chemistry — Structure of Atom]

Two electrons occupying the same orbital are distinguished from one another by which quantum property?

(A) Principal quantum number

(B) Azimuthal quantum number

(C) Magnetic quantum number

(D) Spin quantum number

Correct Answer: (D) Spin quantum number

Explanation: By the Pauli Exclusion Principle, no two electrons in an atom can have identical values for all four quantum numbers. Two electrons sharing the same orbital must differ in spin (+1/2 and -1/2).

Q20. [NEET 2016 — Physics — Magnetism and Matter]

Magnetic susceptibility is negative for which class of materials?

(A) Ferromagnetic

(B) Paramagnetic

(C) Diamagnetic

(D) Antiferromagnetic

Correct Answer: (C) Diamagnetic

Explanation: Diamagnetic substances are weakly repelled by an external magnetic field and acquire a small magnetisation opposite to the applied field, giving them a small negative magnetic susceptibility.

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NEET 2017 — Previous Year Questions

Q21. [NEET 2017 — Biology — Body Fluids and Circulation]

Which combination best explains why adult human red blood cells (RBCs) are enucleate?

(A) They are terminally differentiated cells and losing the nucleus maximises space for haemoglobin

(B) They need the extra space to store more DNA

(C) Enucleation helps RBCs divide faster

(D) RBCs never contained a nucleus at any stage

Correct Answer: (A) They are terminally differentiated cells and losing the nucleus maximises space for haemoglobin

Explanation: RBCs are short-lived, non-dividing somatic cells, so a nucleus for future division is unnecessary; losing it frees up internal volume to pack in more haemoglobin for oxygen transport.

Q22. [NEET 2017 — Biology — Body Fluids and Circulation]

The hepatic portal vein carries blood to the liver from which organs?

(A) Kidneys and lungs

(B) Heart and spleen only

(C) Digestive tract, spleen and pancreas

(D) Skin and skeletal muscle

Correct Answer: (C) Digestive tract, spleen and pancreas

Explanation: Blood absorbing nutrients from the gut is first routed through the hepatic portal vein to the liver, letting the liver process/detoxify absorbed substances before they reach general circulation.

Q23. [NEET 2017 — Biology — Human Reproduction]

The hypothalamic hormone GnRH, essential for reproduction, acts primarily on which gland?

(A) Anterior pituitary

(B) Posterior pituitary

(C) Adrenal cortex

(D) Thyroid gland

Correct Answer: (A) Anterior pituitary

Explanation: GnRH (gonadotropin-releasing hormone) stimulates the anterior pituitary to secrete LH and FSH, which act on the gonads to regulate gametogenesis and sex hormone production.

Q24. [NEET 2017 — Biology — Chemical Coordination and Integration]

Which structure in the human body is considered a 'temporary endocrine gland'?

(A) Thyroid gland

(B) Corpus luteum

(C) Adrenal medulla

(D) Pituitary gland

Correct Answer: (B) Corpus luteum

Explanation: The corpus luteum forms from the ruptured Graafian follicle after ovulation and secretes progesterone; it degenerates within about two weeks unless pregnancy occurs, making it genuinely temporary.

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Q25. [NEET 2017 — Biology — Sexual Reproduction in Flowering Plants]

Which group of plants is known for exhibiting double fertilisation?

(A) Bryophytes

(B) Pteridophytes

(C) Gymnosperms

(D) Angiosperms

Correct Answer: (D) Angiosperms

Explanation: Double fertilisation — one sperm fusing with the egg to form the zygote, the other fusing with the polar nuclei to form the triploid endosperm — is a defining, unique feature of angiosperm reproduction.

Q26. [NEET 2017 — Biology — Photosynthesis]

Phosphoenolpyruvate (PEP) is the primary CO_2 acceptor molecule in which category of plants?

(A) C3 plants

(B) C4 plants

(C) CAM plants only at night

(D) Non-green plants

Correct Answer: (B) C4 plants

Explanation: In C4 plants, PEP (not RuBP) captures atmospheric CO_2 in mesophyll cells via PEP carboxylase, forming a 4-carbon acid later shuttled to bundle sheath cells for the actual Calvin cycle.

Q27. [NEET 2017 — Biology — Morphology of Flowering Plants]

In Bougainvillea, thorns are a modification of which plant structure?

(A) Leaf lamina

(B) Axillary bud (stem)

(C) Root hair

(D) Stipule

Correct Answer: (B) Axillary bud (stem)

Explanation: Bougainvillea's thorns develop from axillary buds turning woody and pointed for defence — a stem modification, often confused with leaf-derived spines in other plants like cacti.

Q28. [NEET 2017 — Biology — Molecular Basis of Inheritance]

The final, unequivocal proof that DNA (rather than protein) is the genetic material came from whose experiments?

(A) Griffith

(B) Avery, MacLeod and McCarty

(C) Hershey and Chase

(D) Meselson and Stahl

Correct Answer: (C) Hershey and Chase

Explanation: Using radioactively labelled bacteriophages, Hershey and Chase showed that only the phage DNA entered the bacterial cell and directed new phage production — clinching evidence for DNA as the genetic material.

Q29. [NEET 2017 — Chemistry — Classification of Elements and Periodicity]

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The inability of the ns2 electron pair in heavier p-block elements to participate in bonding is called what effect?

(A) Lanthanide contraction

(B) Inert pair effect

(C) Common ion effect

(D) Shielding effect

Correct Answer: (B) Inert pair effect

Explanation: The inert pair effect explains why heavier elements of a p-block group (e.g., Tl, Pb, Bi) increasingly favour oxidation states two units lower than the group oxidation state.

Q30. **[NEET 2017 — Chemistry — Coordination Compounds]**

What kind of ligand is EDTA (ethylenediaminetetraacetate ion)?

(A) Monodentate

(B) Bidentate

(C) Tetradentate

(D) Hexadentate

Correct Answer: (D) Hexadentate

Explanation: EDTA4- has six donor atoms (four carboxylate oxygens, two nitrogens) that can all coordinate to a single metal ion simultaneously, making it hexadentate.

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NEET 2018 — Previous Year Questions

Q31. [NEET 2018 — Biology — Biodiversity and Conservation]

Which of the following is NOT considered a method of ex-situ conservation?

(A) Seed banks

(B) Botanical gardens

(C) Cryopreservation

(D) Wildlife sanctuaries

Correct Answer: (D) Wildlife sanctuaries

Explanation: Ex-situ conservation protects species away from their natural habitat (zoos, seed banks, botanical gardens, cryopreservation). Sanctuaries and national parks protect species within their natural habitat — in-situ conservation.

Q32. [NEET 2018 — Biology — Organisms and Populations]

In population ecology, what does the term 'natality' refer to?

(A) Death rate

(B) Birth rate

(C) Emigration rate

(D) Carrying capacity

Correct Answer: (B) Birth rate

Explanation: Natality, along with mortality, immigration and emigration, is one of the four basic parameters used to describe changes in population size over time.

Q33. [NEET 2018 — Biology — Sexual Reproduction in Flowering Plants]

Pollen grains can be stored for several years in liquid nitrogen at approximately what temperature?

(A) -96°C

(B) -196°C

(C) -256°C

(D) -20°C

Correct Answer: (B) -196°C

Explanation: Cryopreservation of pollen in liquid nitrogen at -196°C halts metabolic activity almost entirely, allowing long-term storage of viable genetic material.

Q34. [NEET 2018 — Biology — Neural Control and Coordination]

Nissl's bodies, seen in the cytoplasm of neurons, are mainly composed of what cellular structures?

(A) Golgi vesicles

(B) Rough endoplasmic reticulum and ribosomes

(C) Mitochondria

(D) Lysosomes

Correct Answer: (B) Rough endoplasmic reticulum and ribosomes

Explanation: Nissl bodies are clumps of rough endoplasmic reticulum and free ribosomes in the neuron's cell body, responsible for the high rate of protein synthesis neurons require.

Q35. [NEET 2018 — Biology — Photosynthesis in Higher Plants]

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Oxygen is NOT produced during photosynthesis in which category of photosynthetic organisms?

(A) Green algae

(B) Purple and green sulphur bacteria

(C) Cyanobacteria

(D) Mosses

Correct Answer: (B) Purple and green sulphur bacteria

Explanation: These bacteria use hydrogen sulphide, not water, as the electron donor, releasing sulphur rather than oxygen — key evidence (via Van Niel's hypothesis) that oxygen in ordinary photosynthesis comes from water.

Q36. [NEET 2018 — Biology — Ecosystem]

Which organisms are considered the chief producers in the world's oceans?

(A) Mangroves

(B) Diatoms and other phytoplankton

(C) Coral reefs

(D) Sea grasses

Correct Answer: (B) Diatoms and other phytoplankton

Explanation: Microscopic, free-floating photosynthetic algae (diatoms and other phytoplankton) form the base of most oceanic food chains, contributing the overwhelming majority of marine primary productivity.

Q37. [NEET 2018 — Biology — Morphology of Flowering Plants]

The sweet potato is a modified form of which plant organ?

(A) Adventitious root

(B) Underground stem (tuber)

(C) Rhizome

(D) Corm

Correct Answer: (A) Adventitious root

Explanation: Sweet potato is a classic example of a tuberous (fleshy) adventitious root that stores food, distinguishing it from potato, which is a modified underground stem — a frequently confused pair in NEET Botany.

Q38. [NEET 2018 — Biology — Molecular Basis of Inheritance]

The semi-conservative mode of DNA replication was experimentally demonstrated first in which organism?

(A) Bacteriophage T2

(B) Escherichia coli

(C) Drosophila melanogaster

(D) Pisum sativum

Correct Answer: (B) Escherichia coli

Explanation: Meselson and Stahl's classic density-gradient centrifugation experiment on E. coli confirmed each new DNA double helix retains one parental strand and synthesises one new strand.

Q39. [NEET 2018 — Chemistry — Coordination Compounds]

What are the geometry and magnetic behaviour of the nickel carbonyl complex [Ni(CO)4]?

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(A) Square planar, paramagnetic

(B) Tetrahedral, diamagnetic

(C) Octahedral, paramagnetic

(D) Trigonal bipyramidal, diamagnetic

Correct Answer: (B) Tetrahedral, diamagnetic

Explanation: CO is a strong-field ligand that causes complete pairing of Ni's 3d electrons in this zero-oxidation-state complex, leaving no unpaired electrons and giving a tetrahedral, sp3-hybridised structure.

Q40. **[NEET 2018 — Chemistry — The d- and f-Block Elements]**

What is the geometry of iron pentacarbonyl, Fe(CO)5?

(A) Tetrahedral

(B) Square planar

(C) Trigonal bipyramidal

(D) Octahedral

Correct Answer: (C) Trigonal bipyramidal

Explanation: In Fe(CO)5, iron is sp3d hybridised, giving the five CO ligands a trigonal bipyramidal arrangement around the central iron atom.

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NEET 2019 — Previous Year Questions

Q41. [NEET 2019 — Biology — Cell Structure and Function]

Who first proposed the concept 'Omnis cellula-e cellula' (every cell arises from a pre-existing cell)?

(A) Robert Hooke

(B) Rudolf Virchow

(C) Matthias Schleiden

(D) Theodor Schwann

Correct Answer: (B) Rudolf Virchow

Explanation: Virchow's 1855 dictum overturned the idea of spontaneous generation of cells, establishing that cell division is the sole route by which new cells arise.

Q42. [NEET 2019 — Biology — Genetics and Evolution]

According to Hugo de Vries's mutation theory, how do mutations typically arise?

(A) Gradual and directional

(B) Random, discontinuous (saltatory) and directionless, yet heritable

(C) Only in response to environmental need

(D) Non-heritable but rapid

Correct Answer: (B) Random, discontinuous (saltatory) and directionless, yet heritable

Explanation: De Vries' mutation theory, based on his work on Oenothera lamarckiana, proposed that evolution proceeds via sudden, large, random and heritable changes (mutations) rather than only gradual variation.

Q43. [NEET 2019 — Biology — Chemical Coordination and Integration]

How do steroid hormones typically influence the activities of their target cells?

(A) By binding surface receptors and triggering second messengers

(B) By diffusing in and binding intracellular receptors to regulate gene expression

(C) By directly altering membrane permeability only

(D) By activating G-proteins exclusively

Correct Answer: (B) By diffusing in and binding intracellular receptors to regulate gene expression

Explanation: Being lipid-soluble, steroid (and thyroid) hormones cross the plasma membrane directly and act at the level of transcription, unlike the membrane-receptor mechanism typical of peptide hormones.

Q44. [NEET 2019 — Biology — Principles of Inheritance and Variation]

Using recombination frequency between gene pairs as a measure of genetic distance (to build gene maps) was first proposed by whom?

(A) Thomas Hunt Morgan

(B) Alfred Sturtevant

(C) Walter Sutton

(D) Barbara McClintock

Correct Answer: (B) Alfred Sturtevant

Explanation: Sturtevant used recombination frequencies (in centimorgans) between linked genes to construct the first genetic linkage map, in the fruit fly Drosophila melanogaster.

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Q45. [NEET 2019 — Biology — Microbes in Human Welfare]

Thiobacillus, a group of chemosynthetic bacteria, is helpful in carrying out which biogeochemical process?

(A) Nitrogen fixation

(B) The sulphur cycle

(C) Carbon sequestration

(D) Phosphorus mineralisation

Correct Answer: (B) The sulphur cycle

Explanation: Thiobacillus obtains energy by oxidising inorganic sulphur compounds (like H_2S) to sulphate, playing a key role in returning sulphur to a plant-usable form.

Q46. [NEET 2019 — Biology — Human Reproduction]

Why is colostrum, secreted during the initial days of lactation, essential for a newborn?

(A) It is rich in antibodies (IgA) giving passive immunity

(B) It contains only fats for energy

(C) It has no nutritional value

(D) It prevents lactation from starting too early

Correct Answer: (A) It is rich in antibodies (IgA) giving passive immunity

Explanation: Colostrum's antibody content protects the newborn against infections during the vulnerable early days of life, which is why exclusive breastfeeding is medically recommended.

Q47. [NEET 2019 — Biology — Plant Growth and Development]

Where is photoperiod (day length) perceived in a plant, triggering flowering?

(A) Roots

(B) Leaves

(C) Shoot apex only

(D) Flower buds

Correct Answer: (B) Leaves

Explanation: Although flowering occurs at the shoot apex, it is the leaves that contain the phytochrome-based photoreceptor system perceiving day/night length and sending a signal (florigen) to induce flowering.

Q48. [NEET 2019 — Biology — Sexual Reproduction in Flowering Plants]

A persistent nucellus that remains in the mature seed is known by what name?

(A) Scutellum

(B) Perisperm

(C) Aleurone layer

(D) Epiblast

Correct Answer: (B) Perisperm

Explanation: In most seeds the nucellus is consumed as the embryo/endosperm develops. In a few plants (e.g. black pepper, beet), residual nucellus persists as a food-storing tissue called perisperm.

Q49. [NEET 2019 — Chemistry — Environmental Chemistry / Basic Concepts]

The industrial Haber process is used for large-scale production of which compound?

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(A) Nitric acid

(B) Ammonia

(C) Sulphuric acid

(D) Sodium carbonate

Correct Answer: (B) Ammonia

Explanation: The Haber process synthesises ammonia from nitrogen and hydrogen gas under high pressure and temperature with an iron catalyst — the basis of most nitrogenous fertiliser manufacturing.

Q50. [NEET 2019 — Chemistry — Some Basic Concepts of Chemistry]

How many moles of hydrogen gas are required to produce 20 moles of ammonia via the Haber process?

(A) 10 moles

(B) 20 moles

(C) 30 moles

(D) 40 moles

Correct Answer: (C) 30 moles

Explanation: N2 + 3H2 -> 2NH3 shows a 3:2 mole ratio between H2 and NH3. To produce 20 moles of NH3, you need (3/2) x 20 = 30 moles of H2.

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NEET 2020 — Previous Year Questions

Q51. [NEET 2020 — Biology — Diversity in Living World]

Floridean starch, the storage carbohydrate of red algae, has a structure similar to which other storage polysaccharide?

(A) Cellulose

(B) Glycogen

(C) Amylose

(D) Chitin

Correct Answer: (B) Glycogen

Explanation: Floridean starch is structurally closer to glycogen (highly branched glucose polymer) than to the starch found in green plants — a detail frequently tested when comparing algal groups.

Q52. [NEET 2020 — Biology — Animal Kingdom]

Bilaterally symmetrical, acoelomate animals are exemplified by which phylum?

(A) Aschelminthes

(B) Platyhelminthes

(C) Annelida

(D) Mollusca

Correct Answer: (B) Platyhelminthes

Explanation: Platyhelminthes show bilateral symmetry and triploblastic organisation but lack a true body cavity between the body wall and gut, making them acoelomates.

Q53. [NEET 2020 — Biology — Structural Organisation in Animals]

A cockroach can survive for a few days even after its head is removed. What allows this?

(A) It has a second brain in the abdomen

(B) Its nervous/respiratory systems are not centralised in the head; ganglia and spiracles work along the body

(C) It stops needing oxygen entirely

(D) Cockroaches regenerate a new head within hours

Correct Answer: (B) Its nervous/respiratory systems are not centralised in the head; ganglia and spiracles work along the body

Explanation: Because vital functions like respiration (via spiracles) and basic reflexes don't depend solely on the head, a decapitated cockroach can survive briefly until it eventually dies from dehydration or starvation.

Q54. [NEET 2020 — Biology — Biomolecules]

Which is the most abundant protein found in animals?

(A) Keratin

(B) Collagen

(C) Myosin

(D) Haemoglobin

Correct Answer: (B) Collagen

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Explanation: Collagen makes up roughly a quarter to a third of total protein mass in mammals, forming the structural basis of connective tissue, skin, tendons, cartilage and bone matrix.

Q55. [NEET 2020 — Biology — Respiration in Plants]

How many substrate-level phosphorylation events occur in one turn of the citric acid (Krebs) cycle?

(A) One

(B) Two

(C) Three

(D) Four

Correct Answer: (A) One

Explanation: GTP (equivalent to ATP) is generated once per turn, during conversion of succinyl-CoA to succinate — the only substrate-level phosphorylation step in the cycle.

Q56. [NEET 2020 — Biology — Plant Growth and Development]

Which plant growth regulator, sprayed on sugarcane, increases internode/stem length and boosts yield?

(A) Auxin

(B) Gibberellin (GA3)

(C) Cytokinin

(D) Abscisic acid

Correct Answer: (B) Gibberellin (GA3)

Explanation: Gibberellins promote dramatic stem elongation; spraying GA3 on sugarcane increases stem length (the harvested part), directly increasing commercial yield.

Q57. [NEET 2020 — Biology — Principles of Inheritance and Variation]

Experimental verification of the chromosomal theory of inheritance is credited to which scientist?

(A) Gregor Mendel

(B) Thomas Hunt Morgan

(C) Walter Sutton

(D) Theodor Boveri

Correct Answer: (B) Thomas Hunt Morgan

Explanation: Working with Drosophila melanogaster, Morgan provided the first solid experimental evidence linking specific genes (like the white-eye trait) to specific chromosomes, confirming the chromosomal theory.

Q58. [NEET 2020 — Biology — Biotechnology and Its Applications]

The Bt cotton variety, engineered with a toxin gene from Bacillus thuringiensis, is resistant to which pest?

(A) Aphids

(B) Cotton bollworm (Lepidopteran larvae)

(C) Locusts

(D) Whiteflies

Correct Answer: (B) Cotton bollworm (Lepidopteran larvae)

Explanation: The cry genes from Bacillus thuringiensis code for proteins toxic to specific insect larvae; expressed in Bt cotton, the plant resists bollworm attack without external pesticide sprays.

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Q59. [NEET 2020 — Chemistry — Redox Reactions]

On electrolysis of dilute sulphuric acid using platinum electrodes, what product is obtained at the anode?

(A) Hydrogen gas

(B) Oxygen gas

(C) Sulphur dioxide

(D) Ozone only

Correct Answer: (B) Oxygen gas

Explanation: During electrolysis of dilute H_2SO_4, water is oxidised at the anode (releasing O_2 gas), while H_2 gas is liberated at the cathode — the classic demonstration of water electrolysis.

Q60. [NEET 2020 — Physics — Physics and Measurement]

A screw gauge has a least count of 0.01 mm and 50 divisions on its circular scale. What is the pitch of the screw gauge?

(A) 0.05 mm

(B) 0.1 mm

(C) 0.5 mm

(D) 5 mm

Correct Answer: (C) 0.5 mm

Explanation: Pitch = Least Count x Number of divisions on circular scale = 0.01 mm x 50 = 0.5 mm.

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NEET 2021 — Previous Year Questions

Q61. [NEET 2021 — Biology — Body Fluids and Circulation]

Persons with blood group 'AB' are called 'universal recipients'. What is the reason for this?

(A) Their plasma contains both anti-A and anti-B antibodies

(B) Their plasma contains neither anti-A nor anti-B antibodies

(C) Their RBCs carry no surface antigens at all

(D) They have an unusually high RBC count

Correct Answer: (B) Their plasma contains neither anti-A nor anti-B antibodies

Explanation: 'AB' individuals carry both A and B antigens on their RBCs but produce no antibodies against either, so they can receive blood from donors of any ABO group without triggering agglutination.

Q62. [NEET 2021 — Biology — Cell Structure and Function]

Which organelles are included in the endomembrane system?

(A) Mitochondria and ribosomes

(B) Endoplasmic reticulum, Golgi complex, lysosomes and vacuoles

(C) Chloroplasts and peroxisomes only

(D) Nucleus and centrioles

Correct Answer: (B) Endoplasmic reticulum, Golgi complex, lysosomes and vacuoles

Explanation: The endomembrane system consists of internal membranes coordinating sorting/packaging of macromolecules. Mitochondria (semi-autonomous) and ribosomes (non-membranous) are excluded.

Q63. [NEET 2021 — Biology — Biotechnology: Principles and Processes]

The specific recognition sequence identified by restriction endonucleases, at which they cut DNA, is called what?

(A) An Okazaki fragment

(B) A palindromic nucleotide sequence

(C) A poly-A tail

(D) A TATA box

Correct Answer: (B) A palindromic nucleotide sequence

Explanation: Restriction enzymes bind and cut DNA at short palindromic sequences (sequences reading the same on both strands in the 5'->3' direction).

Q64. [NEET 2021 — Biology — Biotechnology and Its Applications]

Among the pairs 'Drugs-Ricin', 'Alkaloids-Codeine', 'Toxin-Abrin' and 'Lectins-Concanavalin A', which pairing is incorrect?

(A) Drugs-Ricin

(B) Alkaloids-Codeine

(C) Toxin-Abrin

(D) Lectins-Concanavalin A

Correct Answer: (A) Drugs-Ricin

Explanation: Ricin and Abrin are plant-derived toxins, Codeine is an alkaloid drug, and Concanavalin A is a well-known lectin — mixing up 'toxin' and 'drug' is the trap in this question.

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Q65. [NEET 2021 — Biology — Diversity in Living World]

Genera like Selaginella and Salvinia produce two kinds of spores. What is this condition called?

(A) Homosporous

(B) Heterosporous

(C) Isosporous

(D) Monosporous

Correct Answer: (B) Heterosporous

Explanation: Unlike most homosporous pteridophytes, Selaginella and Salvinia produce distinct microspores and megaspores — a trait that foreshadows the seed habit in gymnosperms/angiosperms.

Q66. [NEET 2021 — Biology — Organisms and Populations]

Which factor can lead to the Founder Effect in a newly established population?

(A) Natural selection acting on the whole population

(B) Genetic drift due to the chance gene pool of a small founding group

(C) Uniform gene flow from the parent population

(D) Random mating within a very large population

Correct Answer: (B) Genetic drift due to the chance gene pool of a small founding group

Explanation: Because the founding group is small, its allele frequencies may not represent the source population well; random chance ('drift') can cause the new population to diverge genetically over generations.

Q67. [NEET 2021 — Biology — Morphology of Flowering Plants]

Diadelphous stamens (united into two bundles) are found in which family/example plant?

(A) Solanaceae (e.g., Datura)

(B) Fabaceae (e.g., pea)

(C) Liliaceae (e.g., Lily)

(D) Malvaceae (e.g., China rose)

Correct Answer: (B) Fabaceae (e.g., pea)

Explanation: The (9)+1 diadelphous stamen arrangement is a hallmark floral feature of Fabaceae (pea family), commonly tested alongside vexillary aestivation in the same flower.

Q68. [NEET 2021 — Biology — Diversity in Living World]

In which plant are gemmae (asexual reproductive structures) present?

(A) Funaria (a moss)

(B) Marchantia (a liverwort)

(C) Selaginella

(D) Pteris (a fern)

Correct Answer: (B) Marchantia (a liverwort)

Explanation: Marchantia produces gemmae — small, multicellular vegetative propagules — inside cup-like structures called gemma cups; dispersed by rain, gemmae grow into new plants.

Q69. [NEET 2021 — Chemistry — Chemical Bonding and Molecular Structure]

BF3 is a planar, electron-deficient compound. What are its hybridisation and number of electrons around the central boron atom?

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(A) sp3, 8 electrons

(B) sp2, 6 electrons

(C) sp, 4 electrons

(D) dsp2, 8 electrons

Correct Answer: (B) sp2, 6 electrons

Explanation: Boron in BF3 forms three sigma bonds using sp2 hybrid orbitals, giving a trigonal planar shape; with only 6 electrons around it, BF3 is a classic electron-deficient Lewis acid.

Q70. **[NEET 2021 — Physics — Electrostatics]**

What defining property makes a molecule 'polar'?

(A) It has zero net charge

(B) It possesses a permanent dipole moment from asymmetric charge distribution

(C) It always contains a metal atom

(D) It cannot form hydrogen bonds

Correct Answer: (B) It possesses a permanent dipole moment from asymmetric charge distribution

Explanation: Polar molecules have a net separation of charge even without an external field (e.g., water, HCl), unlike non-polar molecules where the charge distribution is symmetric.

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NEET 2022 — Previous Year Questions

Q71. [NEET 2022 — Biology — Photosynthesis in Higher Plants]

The classic 'girdling' (ringing) experiment, removing a strip of bark from a tree trunk, identifies the pathway of which process?

(A) Upward movement of water through xylem

(B) Downward translocation of food through phloem

(C) Gas exchange through stomata

(D) Absorption of minerals by roots

Correct Answer: (B) Downward translocation of food through phloem

Explanation: Girdling removes the phloem (just under the bark) while leaving xylem intact. Sugar accumulates above the girdle and the region below eventually starves, confirming phloem carries food.

Q72. [NEET 2022 — Biology — Respiration in Plants]

What is the net gain of ATP when one molecule of glucose is converted into two molecules of pyruvic acid during glycolysis?

(A) Net gain of 2 ATP

(B) Net gain of 4 ATP

(C) Net gain of 6 ATP

(D) Net gain of 8 ATP

Correct Answer: (A) Net gain of 2 ATP

Explanation: Glycolysis uses 2 ATP in the investment phase and produces 4 ATP in the payoff phase, giving a net gain of 4 - 2 = 2 ATP per glucose molecule, along with 2 NADH.

Q73. [NEET 2022 — Biology — Principles of Inheritance and Variation]

The XO type of sex determination system is found in which group of organisms?

(A) Birds

(B) Certain insects such as grasshoppers

(C) Humans

(D) Honeybees

Correct Answer: (B) Certain insects such as grasshoppers

Explanation: In the XO system, females are XX while males are X0 (only one sex chromosome) — a variant sex-determination mechanism distinct from human XY and avian ZW systems.

Q74. [NEET 2022 — Biology — Principles of Inheritance and Variation]

During meiosis, at which sub-stage of prophase I do recombination nodules appear on homologous chromosomes?

(A) Leptotene

(B) Zygotene

(C) Pachytene

(D) Diplotene

Correct Answer: (C) Pachytene

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Explanation: Recombination nodules, the sites where crossing over actually occurs between non-sister chromatids, appear during pachytene — after synapsis (zygotene) and before visible chiasmata (diplotene).

Q75. [NEET 2022 — Biology — Excretory Products and their Elimination]

Nitrogenous waste is excreted as a semi-solid pellet or paste by which category of animals?

(A) Ammonotelic animals

(B) Ureotelic animals

(C) Uricotelic animals

(D) Aquatic animals in general

Correct Answer: (C) Uricotelic animals

Explanation: Uricotelic organisms (birds, reptiles, insects) convert nitrogenous waste to relatively insoluble uric acid, excreted as a paste with minimal water loss — critical for animals laying shelled eggs.

Q76. [NEET 2022 — Biology — Morphology of Flowering Plants]

Vexillary aestivation combined with diadelphous stamens is characteristic of which plant?

(A) China rose

(B) Pea (Pisum sativum)

(C) Mustard

(D) Tomato

Correct Answer: (B) Pea (Pisum sativum)

Explanation: In vexillary aestivation, the large 'standard' petal overlaps the two 'wing' petals, which overlap the two 'keel' petals — seen together with diadelphous stamens uniquely in Fabaceae flowers like pea.

Q77. [NEET 2022 — Biology — Plant Growth and Development]

Which plant hormone increases the proportion of female flowers in cucumber, improving fruit yield?

(A) Auxin

(B) Gibberellin

(C) Ethylene

(D) Cytokinin

Correct Answer: (C) Ethylene

Explanation: Ethylene promotes femaleness in flowers of several cucurbits; treating cucumber vines with ethylene increases female flower production and, correspondingly, fruit yield.

Q78. [NEET 2022 — Biology — Microbes in Human Welfare]

Which microorganism produces the immunosuppressive drug cyclosporin A?

(A) Penicillium notatum

(B) Trichoderma polysporum

(C) Monascus purpureus

(D) Clostridium butylicum

Correct Answer: (B) Trichoderma polysporum

Explanation: Cyclosporin A, widely used to prevent organ-transplant rejection, is produced by the fungus Trichoderma polysporum.

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Q79. [NEET 2022 — Chemistry — The p-Block Elements]

What is the IUPAC name assigned to the element with atomic number 119?

(A) Ununoctium

(B) Ununennium

(C) Unbinilium

(D) Ununseptium

Correct Answer: (B) Ununennium

Explanation: IUPAC uses a systematic naming convention based on Latin/Greek roots for the atomic number digits; atomic number 119 (1-1-9) is temporarily called ununennium (symbol Uue).

Q80. [NEET 2022 — Physics — Electromagnetic Induction and Alternating Currents]

The dimensional formula [M L T-2 A-2] corresponds to which physical quantity?

(A) Resistance

(B) Capacitance

(C) Inductance

(D) Magnetic flux

Correct Answer: (C) Inductance

Explanation: Working from V = L(dI/dt), inductance's dimensions combine those of voltage, time and inverse current — [M L T-2 A-2] is the recognised dimensional signature for inductance.

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NEET 2023 — Previous Year Questions

Q81. [NEET 2023 — Biology — Biodiversity and Conservation]

Among the 'Evil Quartet' of causes driving species extinction, which one is considered the most important?

(A) Alien species invasion

(B) Habitat loss and fragmentation

(C) Co-extinction

(D) Over-exploitation

Correct Answer: (B) Habitat loss and fragmentation

Explanation: This exact question — testing the same fact as in the 2016 and 2019 papers — shows how a single high-yield NCERT line can resurface across multiple NEET years.

Q82. [NEET 2023 — Biology — Diversity in Living World]

Which pair of pteridophytes is heterosporous?

(A) Pteris and Adiantum

(B) Selaginella and Salvinia

(C) Equisetum and Lycopodium

(D) Marsilea and Funaria

Correct Answer: (B) Selaginella and Salvinia

Explanation: As in the 2021 question, this recurring question credits Selaginella and Salvinia as the heterosporous pteridophytes named in NCERT.

Q83. [NEET 2023 — Biology — Principles of Inheritance and Variation]

Who first used recombination frequency between gene pairs as a measure of genetic distance to map genes on a chromosome?

(A) Alfred Sturtevant

(B) Thomas Hunt Morgan

(C) Hugo de Vries

(D) William Bateson

Correct Answer: (A) Alfred Sturtevant

Explanation: As in the 2019 question, this recurring question credits Sturtevant with pioneering genetic linkage mapping based on recombination frequency in Drosophila.

Q84. [NEET 2023 — Biology — Ecosystem]

In the equation GPP - R = NPP, what does 'R' represent?

(A) Rate of photosynthesis

(B) Respiration losses of producers

(C) Rainfall

(D) Recycling of nutrients

Correct Answer: (B) Respiration losses of producers

Explanation: Gross Primary Productivity (GPP) is the total rate of organic matter production by photosynthesis; subtracting respiratory losses (R) gives Net Primary Productivity (NPP), available to the next trophic level.

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Q85. [NEET 2023 — Biology — Biotechnology: Principles and Processes]

In the 'gene gun' (biolistic) method of introducing DNA into host cells, microparticles of which metal are typically used?

(A) Silver

(B) Gold (or tungsten)

(C) Copper

(D) Aluminium

Correct Answer: (B) Gold (or tungsten)

Explanation: In the biolistic method, DNA is coated onto microscopic gold or tungsten particles, which are shot into plant cells at high velocity, allowing direct DNA delivery without a biological vector.

Q86. [NEET 2023 — Biology — Respiration in Plants]

Chemiosmosis, the mechanism generating ATP during the electron transport chain, requires which key combination?

(A) A proton gradient across a membrane plus ATP synthase

(B) A calcium gradient plus hexokinase

(C) Substrate-level phosphorylation alone

(D) Only NADH oxidation

Correct Answer: (A) A proton gradient across a membrane plus ATP synthase

Explanation: As protons flow down their concentration gradient through ATP synthase, the enzyme harnesses this flow to catalyse ATP formation — common to mitochondrial respiration and chloroplast photosynthesis.

Q87. [NEET 2023 — Biology — Biomolecules]

Why does cellulose fail to give the characteristic blue-black colour with iodine (unlike starch)?

(A) Cellulose has no glucose units

(B) Cellulose is a linear, unbranched polymer lacking the helical structure needed to trap iodine

(C) Cellulose reacts with iodine to form a colourless product

(D) Cellulose is water-soluble unlike starch

Correct Answer: (B) Cellulose is a linear, unbranched polymer lacking the helical structure needed to trap iodine

Explanation: Starch's amylose component coils into a helix that traps I2 molecules, producing the blue-black colour. Cellulose's straight, rigid chains never form this helical cavity, so no colour develops.

Q88. [NEET 2023 — Biology — Human Health and Disease]

In which type of blood corpuscle does HIV undergo replication to produce progeny viruses?

(A) Red blood cells

(B) B-lymphocytes

(C) Helper T-lymphocytes (TH cells)

(D) Platelets

Correct Answer: (C) Helper T-lymphocytes (TH cells)

Explanation: HIV specifically targets and replicates within helper T-lymphocytes, progressively destroying them; as TH cell numbers collapse, the immune system loses its ability to mount effective defences.

Q89. [NEET 2023 — Chemistry — Equilibrium]

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What is the conjugate base of the Bronsted acid HF?

(A) H3O+

(B) F-

(C) OH-

(D) H2F+

Correct Answer: (B) F-

Explanation: By Bronsted-Lowry theory, an acid's conjugate base is what remains after it donates a proton. Since HF donates H+ to become F-, fluoride ion is HF's conjugate base.

Q90. [NEET 2023 — Physics — Alternating Current]

What is the condition for resonance to occur in a series LCR circuit?

(A) Resistance equals zero

(B) Inductive reactance equals capacitive reactance (XL = XC)

(C) Capacitive reactance is much larger than inductive reactance

(D) Current lags voltage by 90 degrees

Correct Answer: (B) Inductive reactance equals capacitive reactance (XL = XC)

Explanation: At resonance, inductive and capacitive reactances cancel each other exactly, leaving only resistance to oppose current flow — producing maximum current for a given applied voltage.

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NEET 2024 — Previous Year Questions

Q91. [NEET 2024 — Biology — Biomolecules]

Lecithin, a small molecular weight organic compound found widely in living tissues, is an example of which class of biomolecule?

(A) A carbohydrate

(B) A phospholipid

(C) A nucleotide

(D) An amino acid

Correct Answer: (B) A phospholipid

Explanation: Lecithin (phosphatidylcholine) is a major phospholipid component of cell membranes, consisting of glycerol, two fatty acid chains, a phosphate group, and choline.

Q92. [NEET 2024 — Biology — Photosynthesis in Higher Plants]

How many molecules of ATP and NADPH are required to fix one molecule of CO_2 in the Calvin cycle?

(A) 2 ATP and 2 NADPH

(B) 3 ATP and 2 NADPH

(C) 6 ATP and 3 NADPH

(D) 1 ATP and 1 NADPH

Correct Answer: (B) 3 ATP and 2 NADPH

Explanation: Fixing 6 CO_2 molecules (to build one hexose sugar) uses 18 ATP and 12 NADPH in total, which works out to exactly 3 ATP and 2 NADPH consumed per single CO_2 processed through the Calvin cycle.

Q93. [NEET 2024 — Biology — Biotechnology: Principles and Processes]

The restriction enzyme HindII always cuts DNA at a particular recognition sequence. What kind of sequence is this?

(A) A random sequence

(B) A palindromic nucleotide sequence

(C) A poly-T tail

(D) A CpG island

Correct Answer: (B) A palindromic nucleotide sequence

Explanation: HindII was historically the first restriction enzyme shown to recognise and cut a specific palindromic DNA sequence, helping establish the field of recombinant DNA technology.

Q94. [NEET 2024 — Biology — Biomolecules]

What is the cofactor required by the digestive enzyme carboxypeptidase?

(A) Magnesium (Mg^{2+})

(B) Zinc (Zn^{2+})

(C) Iron (Fe^{2+})

(D) Calcium (Ca^{2+})

Correct Answer: (B) Zinc (Zn^{2+})

Explanation: Carboxypeptidase is a zinc-dependent metalloenzyme; the Zn^{2+} ion at its active site is essential for hydrolysing the terminal peptide bond at the carboxyl end of a protein chain.

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Q95. [NEET 2024 — Biology — Plant Growth and Development]

The capacity to generate a whole new plant starting from virtually any living cell of that plant is called what?

(A) Differentiation

(B) Totipotency

(C) Vernalisation

(D) Parthenocarpy

Correct Answer: (B) Totipotency

Explanation: Totipotency is the foundational principle behind plant tissue culture — the idea that even a single differentiated somatic cell retains the full genetic potential to develop into an entire new organism.

Q96. [NEET 2024 — Biology — Biotechnology and Its Applications]

The Bt toxin exists as an inactive protoxin inside Bacillus thuringiensis. What converts it into its active, insect-killing form?

(A) The acidic pH of the mammalian gut

(B) The alkaline pH of the insect's gut

(C) UV light exposure

(D) Heat from cooking

Correct Answer: (B) The alkaline pH of the insect's gut

Explanation: Because the protoxin only becomes active in an alkaline environment (unlike the acidic human gut), Bt crops are considered safe for human consumption while remaining lethal to susceptible insect larvae.

Q97. [NEET 2024 — Biology — Respiration in Plants]

Malonate is a classic inhibitor of which enzyme in the citric acid cycle, and by what mechanism does it act?

(A) Hexokinase, non-competitively

(B) Succinic dehydrogenase, competitively (structural mimic of succinate)

(C) Citrate synthase, irreversibly

(D) Pyruvate dehydrogenase, allosterically

Correct Answer: (B) Succinic dehydrogenase, competitively (structural mimic of succinate)

Explanation: Because malonate structurally mimics succinate, it competes for the enzyme's active site without being converted to product, blocking the succinate-to-fumarate step — a textbook competitive inhibition example.

Q98. [NEET 2024 — Biology — Molecular Basis of Inheritance]

Frederick Griffith's famous experiment on Streptococcus pneumoniae strains demonstrated which fundamental phenomenon?

(A) Conjugation

(B) Transduction

(C) Transformation

(D) Semi-conservative replication

Correct Answer: (C) Transformation

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Explanation: Griffith showed heat-killed virulent (S-strain) bacteria could transfer a 'transforming principle' to living non-virulent (R-strain) bacteria, converting them into virulent bacteria — later identified as DNA.

Q99. [NEET 2024 — Biology — Molecular Basis of Inheritance]

The Meselson and Stahl experiment provided direct evidence for which mode of DNA replication?

(A) Conservative replication

(B) Dispersive replication

(C) Semi-conservative replication

(D) Rolling circle replication

Correct Answer: (C) Semi-conservative replication

Explanation: By tracking DNA density across generations, Meselson and Stahl showed each new DNA molecule contains one old (parental) strand and one newly synthesised strand, confirming semi-conservative replication.

Q100. [NEET 2024 — Biology — Genetics and Evolution]

Which of the following would NOT disturb Hardy-Weinberg equilibrium in a population?

(A) Mutation

(B) Genetic drift

(C) Gene flow

(D) Random mating

Correct Answer: (D) Random mating

Explanation: Hardy-Weinberg equilibrium assumes random mating, no mutation, no migration and no selection. Mutation, drift and gene flow all disturb allele frequencies; random mating is a core assumption, not a disruptive force.

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NEET 2025 — Previous Year Questions

Q101. [NEET 2025 — Biology — Enzymes]

The protein portion of an enzyme (excluding any non-protein cofactor) is called:

(A) Holoenzyme

(B) Apoenzyme

(C) Coenzyme

(D) Prosthetic group

Correct Answer: (B) Apoenzyme

Explanation: An enzyme's protein component alone is the apoenzyme; when combined with its non-protein cofactor (a coenzyme or prosthetic group), the complete, catalytically active molecule is called the holoenzyme.

Q102. [NEET 2025 — Biology — Human Reproduction]

The first menstruation that a girl experiences at puberty is called:

(A) Menopause

(B) Menarche

(C) Ovulation

(D) Parturition

Correct Answer: (B) Menarche

Explanation: Menarche marks the onset of the menstrual cycle at puberty, while menopause refers to its cessation later in life — a classic pair of terms NEET likes to test against each other.

Q103. [NEET 2025 — Biology — Photosynthesis in Higher Plants]

Which of the following statements about the enzyme RuBisCO is true?

(A) It acts only as a carboxylase, never an oxygenase

(B) It is the most abundant enzyme in the world and can act as both a carboxylase and an oxygenase

(C) It functions only in C4 plants

(D) It has no role in the Calvin cycle

Correct Answer: (B) It is the most abundant enzyme in the world and can act as both a carboxylase and an oxygenase

Explanation: RuBisCO (Ribulose bisphosphate carboxylase-oxygenase) is the most abundant enzyme on Earth. Its dual carboxylase/oxygenase activity is exactly what causes photorespiration in C3 plants when CO_2 is limiting.

Q104. [NEET 2025 — Biology — Chemical Coordination and Integration]

Which hormone released from the posterior pituitary is actually synthesised in the hypothalamus?

(A) Growth hormone

(B) Thyroxine

(C) Oxytocin

(D) Insulin

Correct Answer: (C) Oxytocin

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Explanation: Oxytocin and vasopressin (ADH) are synthesised by hypothalamic neurons and simply transported to and released from the posterior pituitary (neurohypophysis), which does not synthesise them itself.

Q105. [NEET 2025 — Biology — Chemical Coordination and Integration]

Why can't insulin be administered orally to diabetic patients?

(A) It is destroyed by stomach acid and proteolytic enzymes before it can act

(B) It causes immediate vomiting

(C) It is not water-soluble

(D) It works only when injected into muscle tissue

Correct Answer: (A) It is destroyed by stomach acid and proteolytic enzymes before it can act

Explanation: Insulin is a peptide hormone; if swallowed, digestive enzymes in the gastrointestinal tract would break it down like any other dietary protein before it could reach the bloodstream intact, so it must be injected.

Q106. [NEET 2025 — Biology — Microbes in Human Welfare]

Streptokinase, produced by the bacterium Streptococcus, is used in medicine for what purpose?

(A) As an antibiotic against bacterial infections

(B) As a 'clot buster' to dissolve blood clots in heart attack patients

(C) As an anti-inflammatory painkiller

(D) As a vaccine adjuvant

Correct Answer: (B) As a 'clot buster' to dissolve blood clots in heart attack patients

Explanation: Streptokinase activates plasminogen to plasmin, which breaks down fibrin clots, making it valuable as a thrombolytic ('clot-busting') drug administered soon after a heart attack.

Q107. [NEET 2025 — Biology — Ecosystem]

Which of the following is used as the unit of productivity of an ecosystem?

(A) Litres per hectare

(B) Grams per square metre per year (g/m2/yr)

(C) Moles per litre

(D) Decibels per second

Correct Answer: (B) Grams per square metre per year (g/m2/yr)

Explanation: Ecosystem productivity — the rate of biomass or energy production — is conventionally expressed as grams of carbon fixed (or dry matter produced) per square metre per year, i.e., g/m2/yr.

Q108. [NEET 2025 — Biology — Molecular Basis of Inheritance]

Which chromosome in the human genome carries the highest number of protein-coding genes?

(A) Chromosome 21

(B) Chromosome Y

(C) Chromosome 1

(D) Chromosome 13

Correct Answer: (C) Chromosome 1

Explanation: Chromosome 1 is both the largest human chromosome and the one carrying the greatest number of genes, a fact highlighted in discussions of the Human Genome Project.

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Q109. [NEET 2025 — Biology — Animal Kingdom]

All living members of class Cyclostomata are best described as:

(A) Ectoparasites on some fishes

(B) Fully independent free-swimming predators

(C) Exclusively freshwater herbivores

(D) Sessile filter feeders

Correct Answer: (A) Ectoparasites on some fishes

Explanation: Cyclostomes (like Petromyzon) are jawless vertebrates that are marine but migrate to fresh water to spawn, and living members are ectoparasitic on other fishes, attaching with a sucking, circular mouth.

Q110. [NEET 2025 — Biology — Cell Cycle and Cell Division]

What is the main function of the spindle fibres during mitosis?

(A) To synthesise new DNA

(B) To pull the separated sister chromatids toward opposite poles of the cell

(C) To break down the nuclear envelope

(D) To supply ATP for cell division

Correct Answer: (B) To pull the separated sister chromatids toward opposite poles of the cell

Explanation: Spindle fibres attach to the kinetochores of chromosomes and contract during anaphase, physically pulling sister chromatids apart toward opposite spindle poles, ensuring each daughter cell gets one full chromosome set.

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Re-NEET 2026 — Previous Year Questions

Q111. [Re-NEET 2026 — Biology — Cell Structure and Function]

The Cell Theory, stating that all living organisms are composed of cells, was formulated by:

(A) Robert Hooke alone

(B) Schleiden and Schwann

(C) Rudolf Virchow alone

(D) Anton van Leeuwenhoek

Correct Answer: (B) Schleiden and Schwann

Explanation: Matthias Schleiden (for plants) and Theodor Schwann (for animals) jointly proposed the Cell Theory in 1838-39; Rudolf Virchow later extended it by explaining that new cells arise from pre-existing cells.

Q112. [Re-NEET 2026 — Biology — Human Health and Disease]

Mad cow disease (bovine spongiform encephalopathy) is caused by which infectious agent?

(A) A virus

(B) A bacterium

(C) A prion (misfolded protein)

(D) A fungus

Correct Answer: (C) A prion (misfolded protein)

Explanation: Prions are infectious, misfolded protein particles with no nucleic acid of their own; they cause other normal proteins to misfold, leading to degenerative neurological diseases like mad cow disease.

Q113. [Re-NEET 2026 — Biology — Morphology of Flowering Plants]

Phyllotaxy refers to the pattern of arrangement of which plant structure on the stem or branch?

(A) Flowers

(B) Leaves

(C) Roots

(D) Fruits

Correct Answer: (B) Leaves

Explanation: Phyllotaxy describes how leaves are arranged around a stem — alternate, opposite, or whorled — an arrangement that maximises each leaf's exposure to light.

Q114. [Re-NEET 2026 — Biology — Diversity in Living World]

In taxonomic classification, the category 'Genus' represents:

(A) A single species only

(B) A group of related species that share more characteristics with one another than with species of other genera

(C) The broadest taxonomic category

(D) A variety within a single species

Correct Answer: (B) A group of related species that share more characteristics with one another than with species of other genera

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Explanation: Genus is a taxonomic rank grouping species that are closely related and share many characteristics, e.g., Panthera includes lion, tiger and leopard as related species within one genus.

Q115. [Re-NEET 2026 — Biology — Breathing and Exchange of Gases]

In water, frogs primarily respire using which structure?

(A) Lungs

(B) Gills

(C) Skin (cutaneous respiration)

(D) Tracheae

Correct Answer: (C) Skin (cutaneous respiration)

Explanation: Submerged frogs rely on cutaneous respiration — gas exchange through their thin, moist, richly vascularised skin — since their lungs cannot function underwater.

Q116. [Re-NEET 2026 — Biology — Locomotion and Movement]

The total number of vertebrae present in the human vertebral column (before any fusion) is:

(A) 26

(B) 33

(C) 24

(D) 30

Correct Answer: (B) 33

Explanation: The human vertebral column develops with 33 vertebrae; in adults, some of these (in the sacral and coccygeal regions) fuse together, reducing the number of separate, movable bones to 26.

Q117. [Re-NEET 2026 — Biology — Diversity in Living World]

A symbiotic association between a fungus and an alga (or cyanobacterium) is called:

(A) Mycorrhiza

(B) Lichen

(C) Coralloid root

(D) Nostoc colony

Correct Answer: (B) Lichen

Explanation: A lichen is the composite organism formed by a fungus (mycobiont) living in close symbiosis with a photosynthetic alga or cyanobacterium (phycobiont), and is sensitive enough to serve as a pollution indicator.

Q118. [Re-NEET 2026 — Biology — Body Fluids and Circulation]

The opening between the right atrium and the right ventricle of the human heart is guarded by which valve?

(A) Bicuspid (mitral) valve

(B) Tricuspid valve

(C) Pulmonary semilunar valve

(D) Aortic semilunar valve

Correct Answer: (B) Tricuspid valve

Explanation: The tricuspid valve, with its three flaps, guards the right atrioventricular opening, while the bicuspid (mitral) valve guards the corresponding opening on the left side of the heart.

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Q119. [Re-NEET 2026 — Biology — Human Reproduction]

Sperm motility (the whip-like swimming movement of sperm) is due to which structure?

(A) The acrosome

(B) The middle piece mitochondria alone

(C) The tail (flagellum)

(D) The head/nucleus

Correct Answer: (C) The tail (flagellum)

Explanation: The sperm's long tail (flagellum) provides the whip-like motion that propels it forward, powered by ATP generated by the mitochondria densely packed in the middle piece just behind the head.

Q120. [Re-NEET 2026 — Biology — Human Reproduction]

Which of the following hormones is NOT secreted by the human placenta?

(A) Human chorionic gonadotropin (hCG)

(B) Human placental lactogen (hPL)

(C) Estrogen and progesterone

(D) Luteinizing hormone (LH)

Correct Answer: (D) Luteinizing hormone (LH)

Explanation: The placenta acts as a temporary but very active endocrine gland, secreting hCG, hPL, estrogens and progesterone — but LH is a pituitary hormone, not one produced by the placenta.

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